

Q1

Observation

Sales

1

10

2

12

3

11

4

13

5

12

6

14

7

50

(i) $\bar{x}$

(ii) $(x_i - \bar{x})^2$

(iii) 6²

(iv) 6

❖ Handling outliers using Z-score method

❖ Apply IQR for outlier handling

12,15,18,19,20,21,22,23,24,80

Observation	Sales
1	10
2	12
3	11
4	13
5	12
6	14
7	50

Lower Bound

Upper Bound

$$\text{Upper Bound} = Q_3 + 1.5 \times \text{IQR}$$

where,
 $\text{IQR} = Q_3 - Q_1$

Any value:

- less than Lower Bound, or
 - greater than Upper Bound
- is considered an outlier.

<i></i> $\bar{x} = 17.43$

<ii> $(x_i - \bar{x})^2$

1. 50.84

2. 26.32

3. 37.58

4. 17.10

5. 26.32

6. 09.80

7. 1080.44

1248.3969

<iii> $\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$

= 178.3424

<iv> $\sigma = 13.35$

<v> Z score $\frac{x_i - \bar{x}}{\sigma}$

1. -0.56

2. -0.40

3. -0.48

4. -0.33

5. -0.406

6. -0.257

7. 2.44

Q IQR

❖ Apply IQR for outlier handling

12, 15, 18, 19, 20, 21, 22, 23, 24, 80

Lower Bound

$$\text{Lower Bound} = Q_1 - 1.5 \times IQR$$

Upper Bound

$$\text{Upper Bound} = Q_3 + 1.5 \times IQR$$

where,

$$IQR = Q_3 - Q_1$$

Any value:

- less than Lower Bound, or
 - greater than Upper Bound
- is considered an outlier.

→ 12, 15, 18, 19, 20, 21, 22, 23, 24, 80

Lower bound 10.5

Upper bound 30.5

Removes 80.

Q2 Remove duplicate values, outliers
& then do min-max normalization.
Fill missing value using mean
SID

Example:

Student ID	Study Hours	Exam Score
S1	2	45
S2	4	60
S3	?	55
S4	6	70
S5	8	85
S6	4	60
S7	10	200

→ Now,

Example:

Student ID	Study Hours	Exam Score
S1	2	45
S2	4	60
S3	?	55
S4	6	70
S5	8	85
S6	4	60
S7	10	200

Fill missing
value with
mean.

So, (5) data
remaining

$$\frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = 6$$

So, Filled Missing

S1	2	45
S2	4	60
S3	6	55
S4	6	70
S5	8	85
S7	10	200

So,	S1	2	45
	S2	4	60
	S3	6	55
	S4	6	70
	S5	8	85

→ Apply min max

$$\frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

S1	0	0
S2	0.33	0.375
S3	0.67	0.25
S4	0.67	0.625
S5	1	1

18/03/26

Sales in Rupees

eg:	X	Y
	01/03/26 10 AM	2000
	02/03/26 2 PM	300
	03/03/26 6 AM	350
	04/03/26 9 AM	400

X					
Date	Day	Weekday/ Weekend	Time	Sales	
01/03/26	Sun	1 (1)	1	200	
02/03/26	Mon	0	2	300	
03/03/26	Tue	0	2	350	
04/03/26	Wed	0	3	400	

Numerical Example – Raw Dataset

eg:

- Suppose we have a dataset of house sales:
- Size (sq.ft): 1000, 1200, 1500
- Bedrooms: 2, 3, 3
- Price (₹ lakhs): 40, 50, 65
- Raw features may not fully capture useful patterns.

ax^2	x^2	x	y
2	1	1	2
9	4	2	8
16	9	3	18
36	16	4	39

$y = ax^2$

• Price per sq feet

$$\begin{array}{l} 1000 \text{ sq feet} \rightarrow 40 \text{ lakh} \\ 1 \text{ sq feet} \rightarrow \frac{40,00,000}{1000} \\ = 4000 \end{array}$$

So, ₹ 4000 per sq feet

$$\begin{array}{l} = \frac{50,00,000}{1200} \\ = 4166.67 \end{array}$$

Similarly,

Q2	Study Hours	Attendance	X_3 (Marks)	Result
	X_1	X_2		
	5	80	35	Fail
	6	70	40	Pass

25/03/26

Q1

x_1	x_2	y
1	1	2
2	2	4
3	3	6

← label/OP/outcome

Q Apply (L1) and (L2)

(i) (L1):

$$TL = MSE + \lambda (|\omega_1| + |\omega_2|)$$

for 1st iteration $\omega_1 = 1, \omega_2 = 1$
 $\lambda = 1$ (for all)

MSE with $\omega_1 = 1, \omega_2 = 1$

$$(a) \quad 1 \cdot 1 + 1 \cdot 1 = 2 = \hat{y}$$

$$y = 2$$

$$\therefore (y - \hat{y})^2 = (2 - 2)^2 = 0$$

$$(b) \quad 2 \cdot 1 + 2 \cdot 1 = 4 \Rightarrow MSE = 0$$

$$(c) \quad 3 \cdot 1 + 3 \cdot 1 = 6 \Rightarrow MSE = 0$$

$$\sum_{i=1}^3 MSE = 0$$

$$\text{Penalty} = \lambda (|1| + |1|)$$

$$= 1(2)$$

$$\text{Penalty} = 2$$

$$\text{T.S} = \text{MSE} + \text{Penalty}$$

$$= 0 + 2$$

$$= 2$$

→ Now, 2nd

$$\omega_1 = 2 \quad \omega_2 = 0$$

$$\text{MSE} = 0$$

$$\text{Penalty} = 2$$

$$\text{T.L} = 2$$

→ 3rd

$$\omega_1 = 1.8 \quad \omega_2 = 0$$

$$\text{MSE} = [(0.2)^2 + (0.4)^2 + (0.6)^2] / 3$$

$$\text{MSE} = \frac{0.56}{3} = 0.1866$$

$$\text{Penalty}^3 = 1.8$$

$$\boxed{\text{T.L} = 1.9866}$$

→ (L_2) :

x_1	x_2	y
1	1	2
2	2	4
3	3	6

$$\begin{array}{l} w_1 : 1 \quad | \quad 2 \quad | \quad 1 \\ w_2 : 1 \quad | \quad 0 \quad | \quad 0.8 \\ (L_2) \end{array}$$

$$T.L = MSE + (|w_1|^2 + |w_2|^2)$$

Solⁿ

1st:

$$MSE = 0$$

$$1+1=2$$

$$\boxed{T.L = 2}$$

2nd : MSE = 0

$$\text{Penalty} = 4$$

$$\boxed{T.L = 4}$$

3rd Iteration = MSE = 0.1866

$$\text{Penalty} = 1.64$$

$$\boxed{T.L = 1.82}$$

02/03/26

PCA → ① Calculate Mean

② Convert dataset wrt mean

③ Find Covariance matrix

④ Find Eigen Value & Eigen Vector

⑤ Obtain Projection Eqn

⑥ Transform dataset wrt Projection

L_1 (Lasso) → $TL = MSE + \lambda (|w_1| + |w_2|)$

L_2 (Ridge) → $TL = MSE + \lambda (w_1^2 + w_2^2)$

Dataset:

x_1	x_2	y
1	6	3
2	1	5
3	1	7

w_1	w_2
2.5	1
3	0
2.0	0
2.5	1
3	0.8
2.6	0.8

(i) Calculate Mean

(ii) Convert dataset wrt mean